

AI SEO Agent - 2026 Edition

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From Optimization to Authority: The New Frontier of Search



Overview

The **AI SEO Agent** is a cutting-edge command-line tool designed to audit and optimize websites for the Generative AI era. Unlike traditional SEO tools that focus solely on keywords and backlinks, this agent evaluates your digital presence against **Generative Engine Optimization (GEO)** standards, **Search Everywhere** strategies, and Google's **E-E-A-T** (Experience, Expertise, Authoritativeness, Trustworthiness) guidelines.

Its mission is simple: **To help your content get cited by AI, not just ranked by algorithms.**

Key Features

-  **Generative Engine Optimization (GEO):** Analyzes content structure (definition lists, tables, data) to maximize "Citation Share" in AI-generated answers (ChatGPT, Gemini, Claude).
-  **Search Everywhere Audit:** Checks for social graph tags and multimodal asset readiness (images, videos) to ensure visibility across all platforms.
-  **E-E-A-T & Authority Checks:** Scans for trust signals, author bylines, and expert validation markers.
-  **Technical SEO Foundation:** Validates critical basics like meta tags, canonicals, and schema markup presence.
-  **Agentic Assets:** Automatically generates modern assets like `suggested_robots.txt` (for AI bot permissions) and `suggested_schema.json`.
-  **Rich CLI Interface:** Features a beautiful, interactive terminal UI with live progress

tracking and color-coded summaries.

Installation

Clone the Repository

```
git clone https://github.com/sgins/SG-SEO.git
```

```
cd SG-SEO
```

1.

Set up the Environment It is recommended to use a virtual environment.

Windows

```
python -m venv venv
```

```
.\venv\Scripts\activate
```

Mac/Linux

```
python3 -m venv venv
```

```
source venv/bin/activate
```

2.

Install Dependencies

```
pip install -r requirements.txt
```

3.

Usage

The agent is run via the command line. The main entry point is `src/main.py`.

Quick Start

Analyze a website (defaults to single page depth):

```
python src/main.py https://www.your-website.com/
```

Advanced Usage

Option	Description	Default
<code>url</code>	The target URL to analyze (required)	N/A
<code>--depth</code>	How deep to crawl internal links	<code>1</code>
<code>--output-dir</code>	Directory to save generated reports	<code>./reports</code>

Example: Deep Crawl To analyze the homepage and all directly linked pages:

```
python src/main.py https://www.your-website.com/ --depth 2
```

Output & Reports

After a successful run, the agent creates a timestamped report folder (default: `reports/`).

1. The Audit Report (`SEO_Report_YYYYMMDD.md`)

A comprehensive Markdown file containing:

- **Executive Summary:** High-level health check.
- **Page-by-Page Analysis:** detailed breakdown of Technical, GEO, and E-E-A-T scores.
- **Strategic Recommendations:** Actionable steps to improve AI visibility.

2. Suggested Assets

- `suggested_robots.txt`: A modernize robots file that explicitly allows helpful AI bots (like GPTBot) to crawl your site, increasing your chances of being cited.
- `suggested_schema.json`: A template JSON-LD structure to help search engines understand your entity (Organization, Website).



Under the Hood

The project is structured for modularity and extensibility:

- `src/main.py`: CLI entry point and UI orchestration.
- `src/crawler.py`: Polite web crawler with depth control.
- `src/analyzer.py`: The brain of the operation. Contains logic for GEO scoring, E-E-A-T detection, and technical checks.
- `src/reporter.py`: Generates the human-readable reports and machine-readable assets.



User Manual: The Educational Guide to Implementation

This section provides comprehensive instructions for executing the SEO and Google Analytics (GA4) strategies across different hosting environments. Before we dive into the commands and clicks, let's establish a shared understanding of the core concepts you will encounter.



Glossary of Core Concepts

- **SEO (Search Engine Optimization)**: The practice of improving a website's technical configuration, content relevance, and link popularity so search engines rank it higher.
- **Google Analytics 4 (GA4)**: The latest version of Google's analytics platform, focusing on event-based tracking rather than pageviews, allowing you to understand user journeys across websites and apps.
- **Google Cloud & "Vibe Coding"**: A custom hosting environment where you have full control over the server and file architecture. "Vibe Coding" refers to writing minimal, effective scripts (like PowerShell) to automate complex tasks rather than relying on bloated software.
- **Wix (Managed CMS)**: A Content Management System where the underlying servers and code are managed for you. You interact with it through a dashboard rather than writing raw code.
- **Sitemap (XML Sitemap)**: A file that lists all the important pages on your website. It acts as a roadmap for search engine crawlers (like Googlebot) so they know exactly what content

exists and when it was last updated.

- **Robots.txt:** A text file placed at the root of a website that gives instructions to web robots (crawlers) about which pages they are allowed to index. "AI-Optimized" means explicitly allowing AI bots (like GPTBot) to read your site.
- **Title Tags & Meta Descriptions:** HTML elements in a page's `<head>`. The Title is the clickable blue link in search results, and the Meta Description is the short summary below it. Both are critical for encouraging users to click.
- **Canonical Tags:** An HTML element that tells search engines which version of a URL is the "master" copy. This prevents penalties if the exact same content is accessible via multiple URLs (e.g., duplicate content).
- **HREFLANG:** An HTML tag used to specify the language and geographical targeting of a webpage. If you have an English version and a Hebrew (`/he/`) version, this tag tells Google which one to show to which user.
- **Google Tag Manager (GTM):** A tag management system that allows you to quickly update tracking codes and related code fragments (tags) on your website without modifying the raw HTML code yourself.
- **BigQuery:** A fully-managed, serverless data warehouse by Google. You can export GA4 data here to run complex SQL queries or feed it into AI models.
- **LLM (Large Language Model):** AI systems like ChatGPT or your local Ollama instance (e.g., Llama 3) that understand and generate human-like text.
- **Schema Markup (JSON-LD):** Structured data vocabulary that helps search engines better understand the info on your website in order to serve "rich results" (like star ratings, prices, or event dates right in the search results).
- **Google Search Console (GSC):** A free Google tool that helps you monitor, maintain, and troubleshoot your site's presence in Google Search results.
- **Core Web Vitals (CWV):** A set of specific factors (LCP, INP, CLS) that Google considers important in a webpage's overall user experience and ranking.
- **Search Intent:** The primary goal a user has when searching a query (e.g., Informational, Commercial Investigation, Transactional).
- **Pillar-Cluster Content:** An SEO architecture where a comprehensive "pillar" page covers a broad topic and links to multiple "cluster" pages that cover specific subtopics in depth.
- **CRO (Conversion Rate Optimization):** The systematic process of increasing the percentage of website visitors who take a desired action (like filling out a lead form).
- **Multi-Touch Attribution:** An analytics approach that assigns value to all the marketing touchpoints a buyer encounters before converting, rather than just the final "last click".
- **MQL (Marketing Qualified Lead):** A prospective customer who has engaged with your content and fits your target buyer profile, signaling they are ready for the next stage in the

sales funnel.

The Theory & Practice of Modern Digital Presence

Before executing the technical scripts, it is crucial to understand the philosophy driving them. Modern web architecture is no longer just about presenting information to humans; it is about structuring data so that machines—specifically search algorithms and AI models—can ingest, understand, and cite your expertise.

1. Search Engine Optimization (SEO): The Theory

At its core, SEO is the science of signaling **relevance** and **authority** to a search engine.

- **The Crawl and Index Phase:** Search engines deploy automated bots (spiders/crawlers) that traverse the web by following links. When a bot arrives at your site, it reads the HTML code and stores a snapshot of your content in a massive database called the "Index." If you are not in the index, you do not exist in search results.
- **The Ranking Phase:** When a user types a query, the search engine sorts through its index to find the most relevant answers. It uses hundreds of ranking signals to do this. Historically, this meant counting how many times a keyword appeared on a page (Keyword Density) or how many other sites linked to yours (Backlinks).
- **The Shift to E-E-A-T & GEO:** Modern SEO has evolved past simple keywords. Google's algorithm now evaluates **E-E-A-T** (Experience, Expertise, Authoritativeness, and Trustworthiness). Furthermore, with the rise of AI, we are entering the era of **Generative Engine Optimization (GEO)**. AI models (like ChatGPT or Google's AI Overviews) synthesize answers directly rather than just providing a list of blue links. To win in GEO, your content must be impeccably structured (using Schema Markup, clear heading hierarchies, and factual density) so that an AI confidently uses you as a primary citation.

2. SEO: The Practice

Translating this theory into practice requires a multi-layered approach:

- **Technical SEO:** This is your foundation. It involves ensuring your site loads in under 2 seconds (Core Web Vitals), possesses a clean `sitemap.xml` so crawlers know where to go, and uses an optimized `robots.txt` to tell bots what they are allowed to index. If the technical foundation is broken, the algorithm abandons the crawl.
- **On-Page SEO:** This is the content layer. It involves optimizing the visible text and the HTML

tags. **Title Tags** and **Meta Descriptions** serve as your digital billboard—they must be highly relevant to the user's search intent. **Canonical tags** prevent the engine from getting confused if two URLs load similar content.

- **Off-Page SEO:** This represents external authority. It involves earning high-quality backlinks from other reputable sites in your industry, which acts as a "vote of confidence" in the eyes of the algorithm.

3. Google Analytics 4 (GA4): The Theory

While SEO drives traffic *to* your site, analytics helps you understand what users do once they *arrive*.

- **The Event-Driven Paradigm:** Previous analytics systems (like Universal Analytics) were "session-based." They tracked pageviews. GA4 is fundamentally different; it is an **event-based** model. Every interaction—a page load, a button click, a video play, a file download, a form submission—is recorded as an "Event." This shift allows for a much more granular understanding of the user journey, especially across different devices (mobile vs. desktop).
- **Privacy & Machine Learning:** In a world where browser cookies are disappearing due to privacy regulations (GDPR, CCPA), GA4 relies heavily on machine learning to "fill in the gaps." It uses behavioral modeling to estimate conversions when users opt out of tracking, ensuring you still have actionable data.

4. GA4 & Data Warehousing: The Practice

Implementing a modern analytics stack involves three major components:

- **The Data Collection Layer (Google Tag Manager):** Instead of writing custom JavaScript for every tracking pixel, you install a single GTM container on your site. You then use the GTM dashboard to fire specific "Tags" (like the GA4 configuration tag) based on specific "Triggers" (like a user clicking the "Start Trial" button).
- **The Reporting Layer (GA4 Dashboard):** This is where you view standard reports—where your traffic is coming from (Acquisition), what pages they view (Engagement), and whether they buy your product (Monetization).
- **The Data Warehouse Layer (BigQuery):** For enterprise systems (like [vectors.co.il](https://www.vectors.co.il)), standard dashboards are often insufficient. By exporting raw GA4 event data to **BigQuery** (Google's SQL data warehouse), you unlock the ability to run advanced analytics, combine web data with your CRM data, or feed user behavioral trends directly into your own local AI

models.

Now that you understand the "why" and "what," let's dive into the "how."

1. vectors.co.il (Google Cloud)

Because this is a custom-hosted environment, we utilize the custom automation scripts provided in the `src/` directory for granular, programmatic control.

A. Generating the Sitemap

As you add new technical content or wiki articles, you need to tell search engines that the site has changed by updating the **Sitemap**.

```
.\src\generate_sitemap.ps1 -SourceDir "C:\path\to\vectors_site\build"
```

This script recursively finds all HTML/MD files, extracts their modification dates, and builds a Google-compliant XML sitemap.

B. Generating an AI-Optimized Robots.txt

To ensure that Generative AI bots (GPTBot, ClaudeBot, etc.) are allowed to ingest your site and cite you as a source, you must define permissions in your **Robots.txt**.

```
.\src\generate_robots.ps1 -OutputDir "C:\path\to\vectors_site\build" -SitemapUrl  
"https://vectors.co.il/sitemap.xml"
```

C. Injecting SEO and GA4 Tags

Instead of manually editing hundreds of HTML files, use this script to automatically inject **Title Tags, Meta Descriptions, Canonical Tags, HREFLANG** localization, and the **Google Tag Manager (GTM)** code directly into your static files.

```
.\src\inject_full_seo_tags.ps1 -TargetDir "C:\path\to\vectors_site\build" -GtmId  
"GTM-XXXXXXX"
```

Note: Replace `GTM-XXXXXXX` with your actual Google Tag Manager ID. The Tag Manager will then

handle forwarding data to **GA4** and eventually to **BigQuery**.

D. Monitoring AI Mentions

To see how AI models categorize your platform, you can use your local **LLM** setup. This script searches for mentions of your brand and asks the AI to analyze the context.

```
python src/ai_mentions_monitor.py --brand "vectors.co.il" --model "llama3"
```

2. ginsbourg.co.il (Wix)

Because Wix is a **Managed CMS**, file-level scripting is not applicable. All integrations must be performed via the Wix Dashboard.

A. SEO Configuration

1. **Initial Setup:** Navigate to **Marketing & SEO > SEO Setup Checklist**. This built-in tool natively connects your site to **Google Search Console (GSC)** so Google can start indexing it.
2. **Schema Markup:** Go to **SEO Settings > Advanced > Structured Data** to input your Service and Organization **JSON-LD**. This will help you get rich results like service snippets in search.
3. **AI Visibility:** Use the Wix AI assistant to monitor and standardize your professional bio across LLMs.

B. Google Analytics 4 Integration

1. Do not manually paste GA4 script codes into the site's custom code section. This can cause duplicate tracking.
2. Navigate to **Marketing & SEO > Marketing Integrations**.
3. Select **Google Analytics** and paste your Measurement ID (**G-XXXXXX**). Wix natively handles the rest.
4. **Enhanced Tracking:** Ensure "Form Interactions" is enabled directly inside your Google Analytics admin panel to automatically track your "Contact Me" inquiries without writing custom event code.

3. Implementing the AI Software Testing SEO Framework

The repository also includes the 10-element framework and 120-day implementation roadmap derived from our *SEO & Google Analytics for AI Software Testing Services* whitepaper (available as [SEO_Analytics_AI_Testing_Whitepaper.md](#)). Here is how to execute it using this tool:

A. Technical & Measurement Foundation (Phase 1)

Ensure your technical foundation is flawless. Run the **AI SEO Agent** to audit your **Core Web Vitals (CWV)** and ensure GA4 and GTM are properly detected:

```
python src/main.py https://your-ai-testing-site.com/ --depth 2
```

Review the agent's output for **Critical** and **Warning** alerts regarding missing Google Tag Manager or GA4 snippets. Proper implementation is essential for enabling **Multi-Touch Attribution** in your reporting.

B. Content Architecture & CRO (Phases 2-4)

Structure your content using the **Pillar-Cluster** model to build topical authority. As you publish new pillar pages and cluster articles, ensure you align landing pages with the appropriate **Search Intent** (e.g., informative clusters vs. transactional pricing pages).

Continually test their performance using **Conversion Rate Optimization (CRO)** techniques. Validate that your content meets Generative Engine Optimization (GEO) standards so that AI models will cite your expertise. Monitor your GA4 Funnel Exploration to identify where users drop off before becoming a **Marketing Qualified Lead (MQL)**.



Future Roadmap

- **Competitor Analysis:** Compare your GEO score against top competitors.
 - **Multimodal Analysis:** Deep scan of images and alt text for accessibility and AI vision understanding.
 - **Agentic Protocol (MCP) Support:** Validation for agent-specific endpoints.
-

Built for the future of search.

SEO Strategic Plan

Comprehensive SEO Strategic Plan: Vectors AI System (vectors.co.il)

This document outlines a multi-phased SEO strategy for **Vectors AI System**, a cutting-edge enterprise cyber-resilience and stress simulation platform. The goal is to establish market authority, capture high-intent search traffic, and drive conversions for both the DIY platform and the Elite Managed Service.

1. Executive Summary

Vectors AI System operates at the intersection of **Performance Engineering** and **Cybersecurity**. While the current landing page is conversion-optimized, the domain lacks the content depth required to rank for competitive industry terms. This plan focuses on building a "Resilience Hub" to attract DevOps, SREs, and C-level executives.

Core Objectives

Objective	Key Metric	Target Date
Increase Organic Visibility	Rank in Top 10 for "Enterprise Load Testing"	Q3 2026
Establish Thought Leadership	12+ High-authority "Cyber-Resilience" guides	Q2 2026
Drive Lead Generation	25% increase in "Start Free Trial" from organic	Q4 2026

2. Technical SEO Foundation

The foundation ensures that search engines can crawl, index, and understand the site's high-performance nature.

Infrastructure & Performance

- **Core Web Vitals:** Maintain LCP (Largest Contentful Paint) under 1.2s. As a performance testing tool, the site *must* be the benchmark for speed.

- **Sitemap & Robots.txt:** Generate a dynamic XML sitemap to ensure new AI-generated test scenarios and blog posts are indexed immediately.
- **Schema Markup:** Implement `SoftwareApplication` and `Service` schema. Specifically, use `Review` and `AggregateRating` for the "Publicly Verified" trust signals.

Internationalization (HREFLANG)

Since the site targets 50+ global regions but currently has a Hebrew (`/he/`) and English version, proper `hreflang` tags are critical to prevent duplicate content issues and ensure users land on the correct regional version.

3. Keyword Strategy & Content Pillars

We will move from simple keyword targeting to **Topic Clusters** centered around cyber-resilience.

Pillar 1: Performance & Load Testing (High Intent)

- **Primary Keywords:** `Enterprise load testing`, `Distributed stress simulation`, `k6 managed service`, `JMeter cloud scaling`.
- **Content Type:** Comparison pages (e.g., "Vectors vs. BlazeMeter"), "How-to" technical documentation, and integration guides for GitHub Actions/Jenkins.

Pillar 2: Cyber-Resilience (Thought Leadership)

- **Primary Keywords:** `Cyber resilience audit`, `Infrastructure stress simulation`, `Adversarial traffic simulation`.
- **Content Type:** Whitepapers on "The ROI of Resilience," case studies on preventing Black Friday crashes, and "State of Cyber-Resilience 2026" reports.

Pillar 3: AI-Driven Testing (Emerging Trends)

- **Primary Keywords:** `AI load test generator`, `Text-to-test scripts`, `Autonomous performance engineering`.
 - **Content Type:** Videos demonstrating the "AI Agent" in action, and landing pages for "Natural Language Load Testing."
-

4. On-Page Optimization Matrix

Element	Current Status	Recommendation
Title Tags	Good	Enhance with years: "Enterprise Cyber-Resilience Platform (2026)"
Meta Descriptions	Generic	Include USP: "Simulate 1M+ users in seconds. No code required. Get your resilience certificate today."
Header Tags (H1-H3)	Present	Optimize H2s for long-tail questions like "How to run a cyber stress test?"
Internal Linking	Limited	Link from "Managed Service" to "Pricing" and "Use Cases" to build topical authority.

5. Off-Page & Authority Building

The "Trust Certificate" Strategy

Leverage the "Publicly Verified" certificates. Encourage clients to host their resilience badges on their own sites, linking back to [vectors.co.il/verify/\[ID\]](https://vectors.co.il/verify/[ID]). This creates a high-authority backlink profile from enterprise-level domains.

Technical Community Engagement

- **Open Source Contributions:** Since the core is built on k6 and JMeter, contribute to these communities and link back to Vectors as the "Scale Layer."
 - **Niche Guest Posting:** Target publications like *DevOps.com*, *The New Stack*, and *Dark Reading*.
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6. Content Roadmap (First 90 Days)

Month 1: The Foundation

- Publish 5 "Integration Guides" (GitHub, GitLab, Jenkins, CircleCI, Azure DevOps).
- Optimize the "Managed Service" landing page for "Performance Architecture Audits."

Month 2: The Comparison Series

- Create "Vectors vs. [Competitor]" pages for the top 5 competitors (BlazeMeter, LoadRunner, k6 Cloud, etc.).
- Launch the "Expert Video Masterclass" series as gated content to capture leads.

Month 3: The AI Narrative

- Publish a deep-dive on "How AI Agents are Replacing Manual Load Scripting."
 - Launch a "Cyber Resilience ROI Calculator" interactive tool.
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7. Monitoring & Analytics

- **Rank Tracking:** Monitor 50+ core keywords across US, EU, and IL markets.
 - **Search Console:** Watch for "Indexation Bloat" given the AI-generated content possibilities.
 - **Conversion Attribution:** Use GA4 to track the journey from "Informational Blog Post" -> "Free Trial" -> "Managed Service Inquiry."
-

Final Note: Vectors AI System has a unique "Managed Service" angle that many SaaS competitors lack. SEO efforts should heavily lean into the "Zero Scripts, Total Performance" value proposition to attract non-technical decision-makers while maintaining technical depth for engineers.

Google Analytics implementation

creating a professional full a. SEO and b. Google Analytics implementation integration plan for both my websites:

1. ginsbourg.co.il on WIX
 2. vectors.co.il on Google Cloud
-

To implement a professional SEO and Google Analytics (GA4) integration for your two platforms, we must address the fundamental architectural differences between a managed CMS like **Wix** and a custom-hosted environment on **Google Cloud**.

Given your focus on performance and medical validation, this plan prioritizes data accuracy and technical speed.

1. ginsbourg.co.il (Wix Implementation)

Wix provides a streamlined but highly automated SEO and tracking infrastructure.

A. SEO Strategy

- **Advanced SEO Setup Checklist:** Access this from your Wix Dashboard. It will automatically connect your site to **Google Search Console**, which is vital for indexing.
- **Structured Data (Schema):** Since you offer consulting services, use the Wix **Schema Markup** tool to add Service and Organization schema. This helps your site appear as a "rich result" in search queries.
- **Core Web Vitals:** Wix now uses Server-Side Rendering (SSR). Minimize the use of heavy third-party Wix Apps, as these can impact your "Good" status in Google's speed metrics.
- **AI Visibility Tool (2026 Update):** Utilize the new Wix AI assistant to track mentions of your services in **Gemini** and **ChatGPT**, ensuring your professional bio is consistent across AI search models.

B. Google Analytics 4 (GA4) Integration

- **Native Integration:** Do **not** manually paste code. Go to *Marketing & SEO > Marketing Integrations* and connect your **Measurement ID** (G-XXXXXXX).
- **Enhanced Measurement:** Ensure "Form Interactions" is enabled in your GA4 admin panel to automatically track your "Contact Me" inquiries without additional coding.

2. vectors.co.il (Google Cloud Implementation)

As a technical architect, you have granular control here. This setup focuses on "Vibe Coding" principles—keeping the core light while using heavy-duty backend tools.

A. SEO Strategy

- **Sitemap & Robots Automation:** Since you manage local LLM inference and a wiki-style structure, ensure your deployment script (Bash/PowerShell) automatically regenerates your sitemap.xml whenever new content is added to your technical folders.
- **Technical Performance:** Leverage **Google Cloud CDN** to serve static assets. Since you are a performance expert, aim for a LCP (Largest Contentful Paint) of under **1.2s**.
- **Canonical Tags:** Use your technical architect access to ensure your /medical/ subdomain correctly uses canonical tags to avoid duplicate content penalties between your main domain and the new subdomain.

B. Google Analytics 4 & BigQuery Integration

For a data-heavy site like vectors.co.il, a simple tracking code is insufficient.

- **Tag Implementation:** Use **Google Tag Manager (GTM)**. This allows you to deploy tracking without redeploying your server-side code on Google Cloud.
- **BigQuery Export:** Enable the GA4-to-BigQuery export. This is essential for your "Agentic Knowledge Core." You can feed this behavioral data back into your local LLMs via Ollama to analyze which technical topics are trending among your visitors.
- **Server-Side Tagging:** Consider setting up a **Cloud Run** instance for Server-Side GTM. This improves privacy compliance and site speed by moving tracking logic off the user's browser.

Implementation Comparison

Feature	Wix (ginsbourg.co.il)	Google Cloud (vectors.co.il)
Setup Method	UI-based / Automated	GTM / Manual Code Injection

Data Storage	Standard GA4 Reports	GA4 + BigQuery Data Warehouse
SEO Control	Managed Templates	Full Programmatic Control
Speed Optimization	Platform-level (SSR)	CDN / Cache / Script Minification
Subdomain Handling	Manual Redirects	Cloud DNS / Virtual Hosting

Critical Next Steps

1. **Consolidate Identity:** Ensure your GA4 properties for both sites use the same **Google Account** to track cross-domain performance.
2. **Verify GSC:** Manually check that both domains are verified in **Google Search Console** to monitor for crawl errors, especially for your new medical validation subdomain.
3. **Monitor "AI Mentions":** Use your local LLM setup to periodically scrape for mentions of "vectors.co.il" to see how AI search engines are categorizing your technical consultancy.

Would you like a specific PowerShell script to automate the sitemap generation for your Google Cloud hosted site?